

Raphael Chia

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Software engineer with 5+ years embedding with client teams and shipping production systems — architected a high-throughput Rust deposit indexer matching against 75M addresses at sub-10 μ s latency with transactional outbox delivery, led Sui integrations end-to-end as technical POC, and shipped a Bittensor trading exchange driving \$4.8M daily volume.

PROFESSIONAL EXPERIENCE

Sui Foundation

Singapore

Solutions Engineer

November 2025–Present

- Embedded with a partner team as technical POC for a Sui-native payment integration — led discovery, shaped architecture, and shipped the full solution alongside their engineers from requirements through production.
- As technical POC for the indexer portion of a partner payment solution, led requirements gathering, architected and built a Rust deposit indexer (sui-indexer-alt-framework, gRPC) matching deposits against **75M wallet addresses** in a Kafka-sourced in-memory HashSet (~**27 GB RAM**), and presented architecture and benchmarks for stakeholder greenlight. Designed a transactional outbox decoupling ingestion from Kafka latency with at-least-once delivery. Benchmarked **500-key lookups at 7.3 μ s** (1,400 $\times$ under gate); cold-boot at **121K msg/s**. Validated end-to-end on testnet.
- Led bootcamps (**30–80 participants**) and hackathons (**up to 500 participants**) as hands-on technical instructor, onboarding builder teams into Sui integration patterns.

Tensorplex Labs

Singapore

Software Engineer, Full Stack

May 2024–November 2025

- Developed and shipped the frontend of **backprop.finance**, a Bittensor trading exchange that scaled from **\$5–8M to \$50–60M weekly volume (36% of network volume)**.
- Profiled and optimized backprop.finance’s input-heavy UI using React concurrent APIs (startTransition, useDeferredValue), component flattening, and re-render elimination to resolve client blocking under high data throughput.
- Actively monitored a **21k-member Discord** to surface pain points and feature requests; shipped referral and leaderboard systems (**800+ wallets**) and a GA-driven usage pipeline that flagged 2 features at **0.1% engagement**, driving data-backed reprioritization and UX redesigns.
- Built an LLM-powered newsletter (DeepSeek-V3 + FastAPI) summarizing weekly X posts, saving **2–4h/week** in manual curation at **23% engagement**; created an internal LLM labeling platform (FastAPI + Next.js) cutting review time from **>2 min to <30 sec**.
- Reduced trading-metrics API latency from **15s to <0.5s** via Redis caching, chunked processing, and response streamlining.

Telepathy Labs

Singapore

DevOps Engineer

Nov 2022–May 2024

- Restructured CI/CD for parallel develop-and-test workflows; automated cross-team service versioning and release tracking.

NCS

Singapore

Software Developer

Jan 2021–Nov 2022

- Built a document-overdue alerting module for Singapore’s Attorney General Chamber and a digital lobby-onboarding app, reducing deadline-related errors and manual entry processing.

EDUCATION

SINGAPORE INSTITUTE OF TECHNOLOGY

Singapore

Bachelor of Engineering in Infocomm Tech, Software Engineering

September 2022

SKILLS

- Systems & Infra:** Rust, Kafka, PostgreSQL, Redis, RocksDB, Docker, gRPC, EBS/PVC, Prometheus
- Web & App:** TypeScript, React, Next.js, Node.js, Prisma, FastAPI, Python
- Domain:** Blockchain indexing, Move smart contracts, high-throughput pipeline design, transactional outbox patterns